

InternNova Data Analytics Internship

Week 5 Assignment: Power BI & Business Intelligence

Submitted by: Brojo Mohan Dutta

Duration: 1 Week (7 Days) | Total Marks: 100

Power BI data preparation steps

Data model screenshots

DAX formulas and outputs

Dashboard screenshots

Insights and recommendations

Task 1: Power BI Introduction & Data Import

1.1 Purpose of Power BI in Data Analytics

Power BI is a business intelligence platform that connects to raw data sources, transforms and cleans that data through Power Query, models relationships between tables, and converts the result into interactive visuals and dashboards. This enables data-driven decision-making without requiring end users to write code.

1.2 Dataset Overview

The project uses a sales dataset containing transactional sales information across two related tables.

Primary transaction table: `sales_fact` (June 2024 – September 2024)

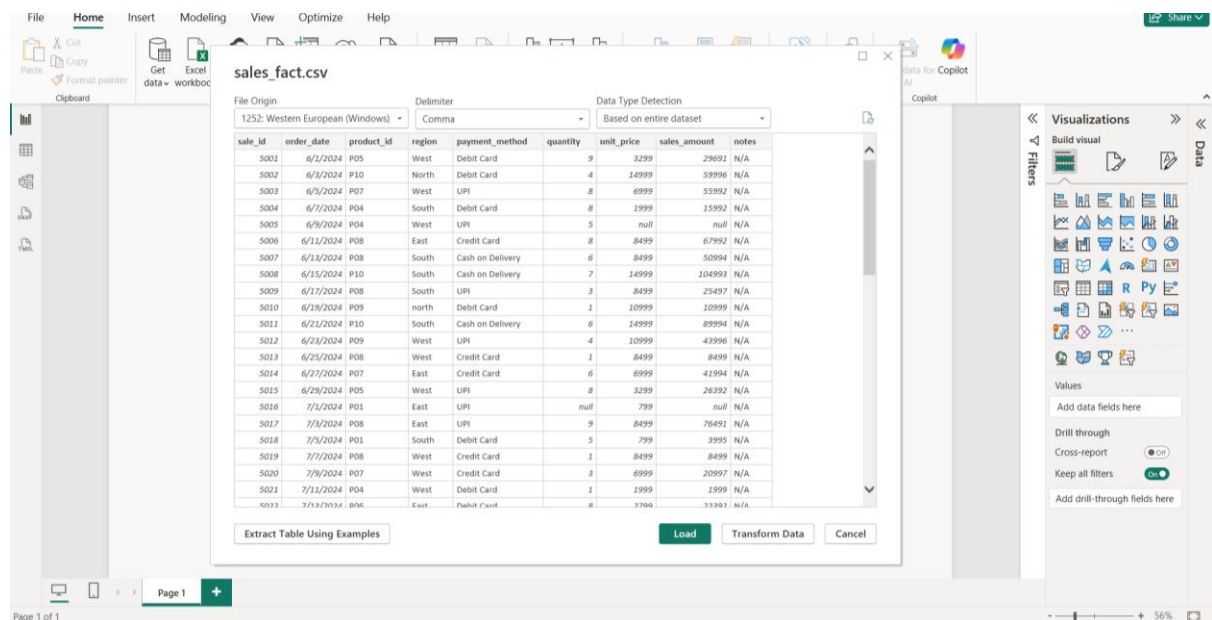
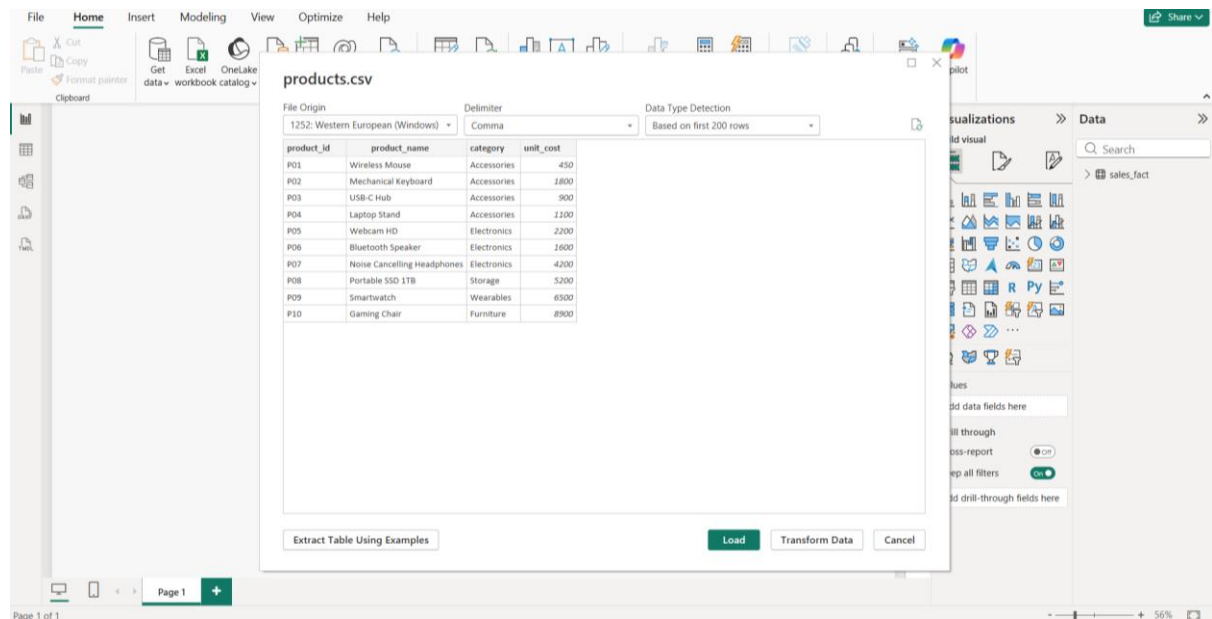
Field	Description
<code>sale_id</code>	Unique transaction identifier
<code>order_date</code>	Date of transaction
<code>product_id</code>	Linked product identifier
<code>region</code>	Sales region
<code>payment_method</code>	Payment channel used
<code>quantity</code>	Units sold
<code>unit_price</code>	Price per unit
<code>sales_amount</code>	Transaction revenue
<code>notes</code>	Placeholder field (removed during cleaning)

Product dimension table: `products`

Field	Description
<code>product_id</code>	Unique product identifier
<code>product_name</code>	Name of the product
<code>category</code>	Product category
<code>unit_cost</code>	Cost per unit to the business

1.3 Data Import

Both datasets were imported into Power BI Desktop via **Get Data** → **Text/CSV**, and inspected in the **Table view** before proceeding to Power Query for cleaning and preparation.



Task 2: Data Cleaning & Preparation

2.1 Objective

Power Query was used to inspect and transform the imported data, ensuring it was consistent, correctly typed, and suitable for accurate analysis.

2.2 Data Preparation Steps

#	Step	Detail
1	Promoted Headers	The first row of raw data was promoted to become the table's column headers.
2	Data Type Correction	order_date → Date; quantity → Whole Number; unit_price → Fixed Decimal Number; sales_amount → Fixed Decimal Number.
3	Removed Unnecessary Column	The notes column was removed — it contained only placeholder "N/A" text with no analytical value.
4	Text Standardization	The region column was standardized using Transform → Format → Capitalize Each Word , correcting inconsistent casing ("north" → "North", "SOUTH" → "South").
5	Missing Value Handling	Null values in quantity and unit_price were replaced using Replace Values : quantity nulls filled with the column median (5); unit_price nulls filled with the product's typical price (₹1,999 for product P04).
6	Incorrect Value Correction	A negative quantity value (-3, sale_id 5031) was identified as a data entry error and corrected to 3 via Replace Values.
7	Recomputed Sales Amount	The original sales_amount column was removed and replaced with a custom column: [Quantity] * [Unit Price], ensuring consistency after the missing-value fixes above.
8	Removed Duplicates	A duplicate transaction (sale_id 5008) was identified and removed via Remove Rows → Remove Duplicates .
9	Renamed Columns	Columns were renamed to Title Case for report clarity (e.g., sale_id → "Sale ID", order_date → "Order Date"), applied consistently across both tables.

10	Applied Changes	Close & Apply was used to load the cleaned tables into the Power BI data model.
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Query Settings

PROPERTIES

Name

sales_fact

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed 'order_date' Type to Date

Changed 'unit_price' to Currency

Changed 'quantity' to Whole number

Removed 'notes' Column

Standardized 'region' column

Handled Missing Quantity Values

Corrected Missing Unit Price

Corrected Negative Quantity

Removed Original 'sales_amount' Column

Added Custom Calculated 'Sales Amount' Column

Formatted Sales Amount as Currency

Removed Duplicate Record

Renamed Columns

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Table: sales_fact (60 rows)

Sale ID	Order Date	Product ID	Region	Payment Method	Quantity	Unit Price	Sales Amount
5001	01/06/2024	P05	West	Debit Card	9	3,299	29,691
5002	03/06/2024	P10	North	Debit Card	4	14,999	59,996
5003	05/06/2024	P07	West	UPI	8	6,999	55,992
5004	07/06/2024	P04	South	Debit Card	8	1,999	15,992
5005	09/06/2024	P04	West	UPI	5	1,999	9,995
5006	11/06/2024	P08	East	Credit Card	8	8,499	67,992
5007	13/06/2024	P08	South	Cash on Delivery	6	8,499	50,994
5008	15/06/2024	P10	South	Cash on Delivery	7	14,999	104,993
5009	17/06/2024	P08	South	UPI	3	8,499	25,497
5010	19/06/2024	P09	North	Debit Card	1	10,999	10,999
5011	21/06/2024	P10	South	Cash on Delivery	6	14,999	89,994
5012	23/06/2024	P09	West	UPI	4	10,999	43,996
5013	25/06/2024	P08	West	Credit Card	1	8,499	8,499
5014	27/06/2024	P07	East	Credit Card	6	6,999	41,994
5015	29/06/2024	P05	West	UPI	8	3,299	26,392
5016	01/07/2024	P01	East	UPI	5	799	3,995
5017	03/07/2024	P08	East	UPI	9	8,499	76,491
5018	05/07/2024	P01	South	Debit Card	5	799	3,995
5019	07/07/2024	P08	West	Credit Card	1	8,499	8,499
5020	09/07/2024	P07	West	Credit Card	3	6,999	20,997
5021	11/07/2024	P04	West	Debit Card	1	1,999	1,999
5022	13/07/2024	P06	East	Debit Card	8	2,799	22,392
5023	15/07/2024	P09	South	Debit Card	7	10,999	76,993
5024	17/07/2024	P09	North	Credit Card	3	10,999	32,997
5025	19/07/2024	P08	North	Debit Card	6	8,499	50,994
5026	21/07/2024	P06	South	Credit Card	2	2,799	5,598
5027	23/07/2024	P01	West	Debit Card	1	799	799
5028	25/07/2024	P01	South	Cash on Delivery	1	799	799

2.3 Purpose of Data Cleaning

Incorrect data types, missing values, inconsistent text, and unnecessary fields can lead to inaccurate calculations and misleading visualizations. This cleaning process ensured the dataset was analytically reliable before modeling began.

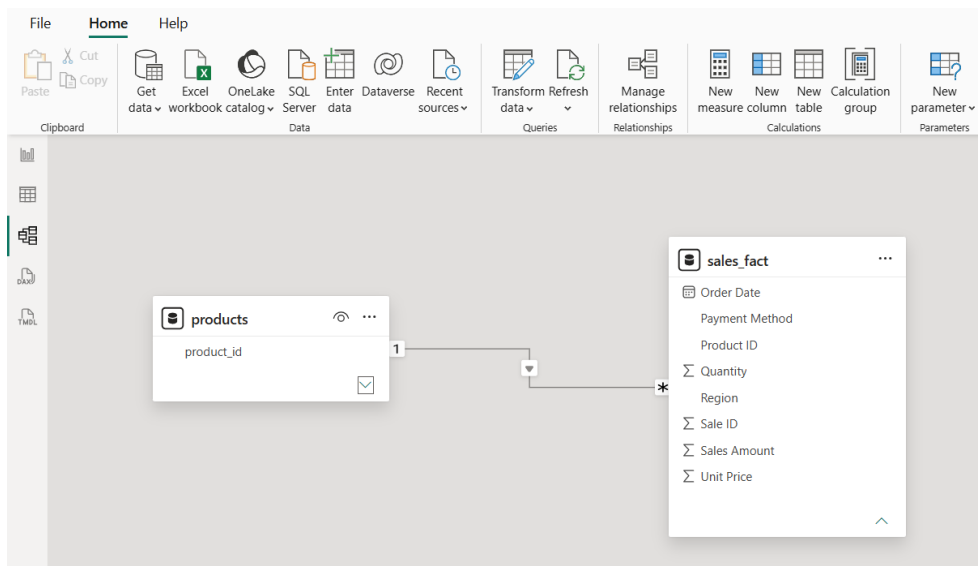
Task 3: Data Modeling & Relationships

3.1 Relationship Setup

The common column identified between the two tables was **product_id**.

Step	Action
1	Opened Model view in Power BI Desktop.
2	Dragged product_id from sales_fact onto product_id in products (alternatively via Modeling → Manage Relationships → New).
3	Set Cardinality : One-to-Many (1:*) — products [product_id] is unique (One side); sales_fact [product_id] repeats (Many side).

4	Set Cross-filter direction : Single (filters flow from <code>products</code> → <code>sales_fact</code>).
5	Confirmed relationship set to Active .



3.2 Role of Each Table

products — descriptive dimension table

- Product ID, Product Name, Category, Unit Cost

sales_fact — transactional fact table

- Sale ID, Order Date, Product ID, Region, Payment Method, Quantity, Unit Price, Sales Amount

3.3 Explanation of the Model

This One-to-Many relationship allows a single product record to relate to many sales transactions, enabling Power BI to analyze sales by product, category, region, and other business dimensions — while filters applied on the `products` table (e.g., by category) automatically propagate to filter `sales_fact`.

Task 4: DAX Calculated Columns & Measures

4.1 Calculated Columns

Total Cost (transaction-level cost)

The total cost for each transaction was calculated using quantity and the corresponding product unit cost.

```
Total Cost = sales_fact[Quantity] * RELATED(products[unit_cost])
```

Uses `RELATED()` to pull the correct unit cost from the `products` table via the model relationship.

Profit (transaction-level profit)

Profit was calculated as the difference between sales revenue and total product cost.

```
Profit = sales_fact[Sales Amount] - sales_fact[Total Cost]
```

This provides a transaction-level profit value that can subsequently be aggregated using measures.

Month (for time-based trend analysis)

The month extracted from Order Date for monthly sales trend visual.

```
Month = FORMAT(sales_fact[Order Date], "MMM")
```

4.2 Measures

Measure	DAX Formula	Dashboard Result
Total Sales Amount	<code>SUM(sales_fact[Sales Amount])</code>	≈ ₹1.50M
Total Profit	<code>SUM(sales_fact[Profit])</code>	≈ ₹606.18K
Total Transactions	<code>COUNTROWS(sales_fact)</code>	60
Average Sale Value	<code>AVERAGE(sales_fact[Sales Amount])</code>	≈ ₹25.03K
Profit Margin	<code>DIVIDE([Total Profit], [Total Sales Amount], 0)</code>	≈ 40.36%



1 Total Sales Amount = `SUM(sales_fact[Sales Amount])`



1 Total Profit = `SUM(sales_fact[Profit])`



1 Total Transactions = `COUNTROWS(sales_fact)`



1 Average Sale Value = `AVERAGE(sales_fact[Sales Amount])`



1 Profit Margin = `DIVIDE([Total Profit], [Total Sales Amount], 0)`

4.3 Summary of DAX Calculations

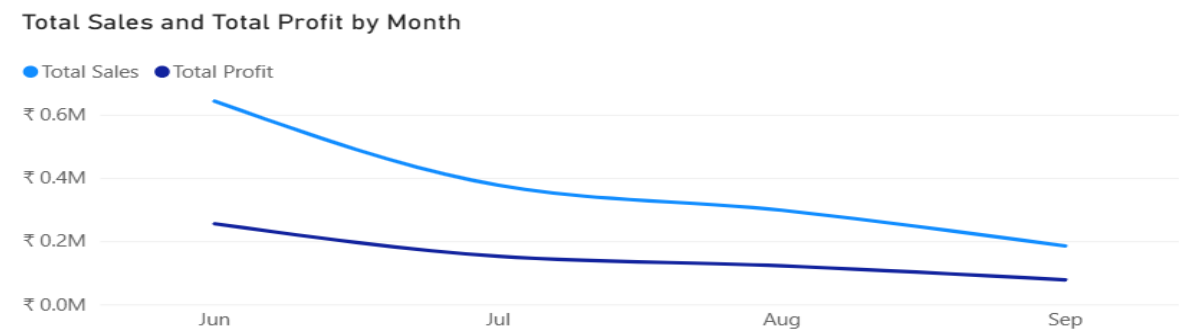
Calculation	Purpose
Total Cost	Calculates transaction-level cost
Profit	Calculates transaction-level profit
Total Sales Amount	Calculates total revenue
Total Profit	Calculates total profit
Total Transactions	Counts unique transactions
Average Sale Value	Calculates average transaction value
Profit Margin	Measures profitability as a percentage of sales

Task 5: Power BI Visualizations

5.1 Monthly Sales Trend — Line Chart

Fields: X-axis → Month | Y-axis → [Total Sales Amount], [Total Profit]

Title: *Monthly Sales Trend*

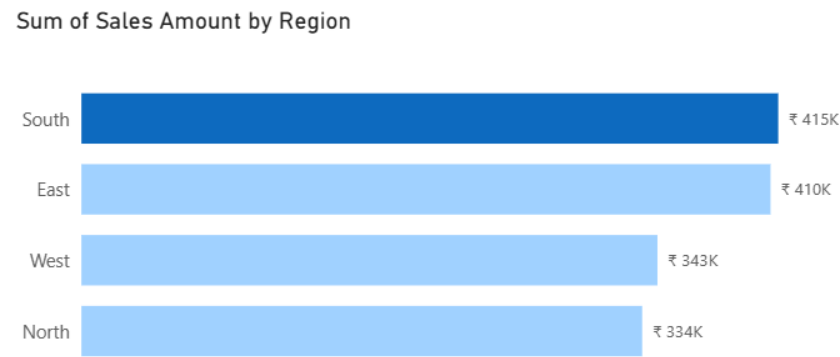


Month	Total Sales
June	₹668.02K
July	₹403.00K
August	₹397.70K
September	₹184.54K

The chart reveals a substantial and consistent decline in sales across the analysis period.

5.2 Sales by Region — Bar Chart

Fields: Y-axis → Region | X-axis → [Total Sales Amount]
Title: Sales by Region

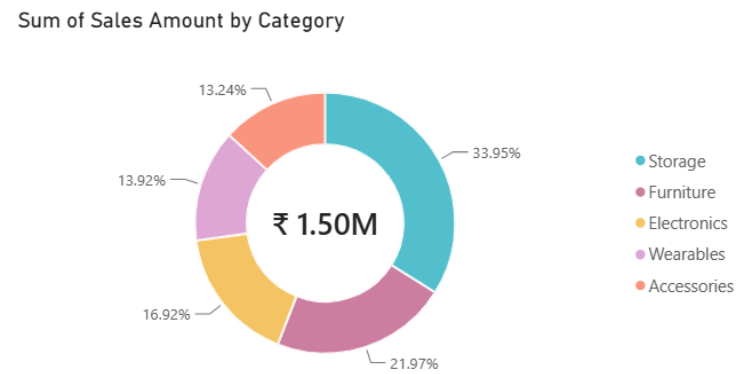


Region	Approx. Sales
South	₹415K
East	₹410K
West	₹343K
North	₹334K

South is the strongest-performing region, closely followed by East, while North trails all regions.

5.3 Sales by Category — Donut Chart

Fields: Legend → products[Category] | Values → [Total Sales Amount]
Title: Sales by Category



Category	Share of Sales
Storage	34%
Furniture	22%
Electronics	17%
Wearables	14%
Accessories	13%

Storage is the leading category, contributing roughly one-third of total sales.

5.4 Category × Region Performance — Matrix

Fields: Rows → products[Category] | Columns → sales_fact[Region] | Values → [Total Sales Amount]

Title: Category × Region Performance Matrix

CATEGORY × REGION PERFORMANCE MATRIX					
Region	Accessories	Electronics	Furniture	Storage	Wearables
East	₹ 3,995.00	₹ 102,279.00	₹ 14,999.00	₹ 288,966.00	
North	₹ 92,358.00	₹ 6,999.00	₹ 74,995.00	₹ 93,489.00	₹ 65,994.00
South	₹ 48,973.00	₹ 8,897.00	₹ 194,987.00	₹ 84,990.00	₹ 76,993.00
West	₹ 53,560.00	₹ 135,871.00	₹ 44,997.00	₹ 42,495.00	₹ 65,994.00

Notable high-performing combinations:

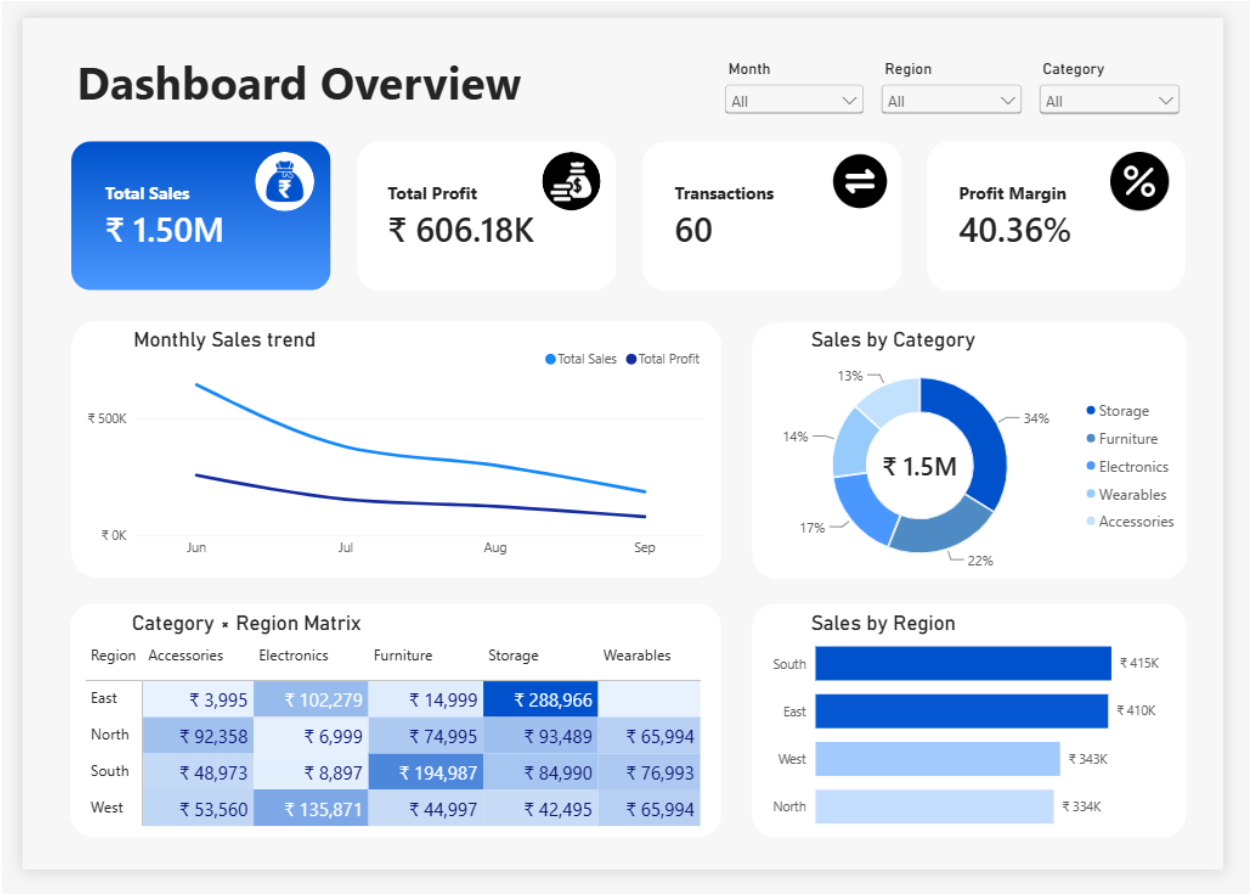
- Storage in East — ≈ ₹288,966
- Furniture in South — ≈ ₹194,987
- Electronics in West — ≈ ₹135,871

Conditional formatting (color scales) was applied to make high and low values easier to identify at a glance.

Task 6: Interactive Dashboard & Filters

6.1 Dashboard Overview

The final deliverable is a single-page **Sales Performance Dashboard**, combining KPI cards, charts, a matrix, and slicers into a concise, interactive view of business performance.



6.2 KPI Cards

KPI	Result
Total Sales	₹1.50M
Total Profit	₹606.18K
Total Transactions	60
Profit Margin	40.36%

6.3 Visualizations Included

- Monthly Sales Trend (Line Chart)
- Sales by Region (Bar Chart)
- Sales by Category (Donut Chart)
- Category × Region Performance (Matrix)

6.4 Interactive Slicers

Slicer	Function
Month	Filters analysis to a specific month
Region	Focuses analysis on a specific geographic region
Category	Filters by individual product category

These slicers make the report fully interactive, allowing end users to explore the data dynamically without modifying the underlying dataset.

6.5 Layout Summary

KPI cards are arranged in a single row at the top for an immediate performance snapshot, slicers are placed top-right for easy access, and the four supporting visuals plus the matrix are arranged below in a clean, evenly-spaced grid — providing a logical flow from summary → trend → breakdown → detail.

Task 7: Business Insights & Recommendations

7.1 Key Insights

- Sales declined sharply and consistently across every month of the analysis period.** Revenue fell from ₹668.02K in June to ₹184.54K in September — a drop of roughly 72% from peak to trough — making this the single most urgent trend on the dashboard.
- Storage is the top-performing product category**, contributing 34% of total sales share, ahead of Furniture (22%) and more than double Electronics (17%). The Storage–East combination alone (≈₹288,966) is the single strongest category-region pairing in the matrix.
- South is the leading sales region**, generating ≈₹415K, narrowly ahead of East (≈₹410K), while North is the weakest performer at ≈₹334K — a gap of roughly 19% between the top and bottom regions.
- Overall profitability is strong, with a 40.36% profit margin** (≈₹606.18K profit on ≈₹1.50M in sales) — indicating that despite the revenue decline, the underlying unit economics of the business remain healthy.
- Sales performance is relatively concentrated in the South and East regions.** South generated approximately **₹415K**, followed closely by East at approximately **₹410K**. Together, these two regions account for a substantial share of total revenue, while North records the lowest regional sales at approximately ₹334K. This indicates that South and East are currently the strongest markets and should receive attention for inventory availability and customer retention.

7.2 Business Recommendations

Recommendation 1 — Investigate and reverse the June–September revenue decline.

Every month in the dataset shows lower sales than the one before it, culminating in a 72% drop from peak to trough. Given the dashboard's otherwise healthy 40.36% profit margin, this decline is a volume/demand problem rather than a pricing or cost problem. **Action:** cross-reference monthly unit counts against marketing activity, stock availability, and seasonality to isolate the root cause before revenue erodes further.

Recommendation 2 — Grow the underperforming Accessories and Wearables categories through cross-selling.

Accessories (13%) and Wearables (14%) are the two smallest categories on the donut chart, together contributing less than half of Storage's share alone. **Action:** bundle these lower-ticket items with top sellers — e.g., offer a discounted Accessory or Wearable add-on with every Storage or Furniture purchase — to lift their visibility and unit volume without requiring a standalone marketing push.

Recommendation 3 — Replicate the Electronics-in-West success pattern in other regions.

The Category × Region matrix shows Electronics in West performing strongly ($\approx ₹135,871$), noticeably ahead of Electronics' performance elsewhere, while West as a whole still trails South and East overall. **Action:** analyze what's driving Electronics' strength specifically in West (local demand, pricing, promotions) and test applying the same approach to Electronics sales in North and East, where the category's contribution is comparatively weaker — a targeted way to lift underperforming regions without changing the whole regional strategy.
